

DB Tutorial

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Last Update: September 10, 2026

1 Findings

1.1 Move toward ground state

I had an interesting observation regarding DB-QITE: The purpose of this algorithm is that when applied to state $|\psi_{old}\rangle$ we get $|\psi_{new}\rangle$ that is closer to the ground state of H than before.

To measure how close a state ($|\psi\rangle$) is to the ground state ($|v_0\rangle$) we can use the size of dot product: $|\langle v_0|\psi\rangle|$. So the purpose of DB-QITE is to ensure

$$|\langle v_0|\psi_{new}\rangle| > |\langle v_0|\psi_{old}\rangle|$$

My observation is that if we show $|\psi_{old}\rangle = (a_0, a_1, \dots, a_n)$ in the basis of H eigenvectors, we will have:

$$|\langle v_0|\psi_{new}\rangle|^2 = |\langle v_0|\psi_{old}\rangle|^2 + t\left(\sum_{i=0}^n 2(s_i - s_0)|a_i|^2|a_0|^2\right) + O(t^2)$$

where s_i is the i 'th smallest eigenvalue of H and t is time.

It is obvious that the value added to $|\langle v_0|\psi_{old}\rangle|^2$ is always positive except for when $|a_0| = 0$ or 1 therefore we always get closer to the ground state. Don't worry about the two edge cases. They mean $|\psi_{old}\rangle$ has no ground state component or is already ground state itself, so for the former convergence never happens using ITE algorithms and for the latter we have already converged.

Intuition: We can see that change in $|\langle v_0|\psi\rangle|^2$ is proportional to $\sum_{i=0}^n (s_i - s_0)|a_i|^2$. This sum actually has an interesting physical interpretation. It is equal to $\langle\psi|H|\psi\rangle - s_0$ which means how far the average energy of the state is from the ground state. This says that for the same starting $|\langle v_0|\psi\rangle|$, the states with higher average energy, converge faster towards the ground state.

Details and proof can be found in [here](#)